

Modeling Safer Outcomes: Predicting and Preventing Severe Crash Injury

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1 Executive Summary

Motor vehicle crashes in Massachusetts represent a significant and measurable public health and economic risk. Using statewide motor vehicle crash data from 2022- 2023 obtained from the MassDOT Crash Data Portal. This project applies an actuarial risk modeling framework to identify the structural, behavioral and environmental factors that most affect crash injury severity. The dataset include 251,363 police reported crashes and over 100 exploratory variables such as roadway design, driver characteristics, environmental factors, and crash dynamics.

Crash severity was modeling using an Ordered Logistic Regression model. The model estimates the probability of increasing severe outcome conditional effect on crash occurrence. Severity was defined on a 4 level scale: property damage, minor injury, serious injury and fatal injury. This modeling approach preserves the ordered structure of injury outcomes and provides interoperable risk multipliers through odd ratios. The model archives an out of sample prediction accuracy of around 75.6%, this indicates strong predictive performance.

Result show than severe injury outcomes are not random but are strongly associated with identifiable and quantifiable risk factors, The largest predictors of highest severity include driver medical illness (odds ratio = 4.45), speed related driving behavior (odd ratio =2.66), driver fatigue (odd ratio = 1.75), and intersection presence (around 33% higher odds of severe injury). Structural roadway characteristics also significantly influence these outcomes. Higher speed environments and multi vehicle crashes increase severity risk , while controlled access design reduces severity by limiting conflicts. Night time conditions also increase injury severity risk, while wet surface conditions have a lower severity, probably because of reduced travel speed.

Risk characterization shows that fatal crashes represent less than 1% of incidents. These crashes represent disproportionately large social and economic losses. Using excepted value principles, even small reductions in severe injury probability produce substantial economic benefit. The model based simulation indicated that removing intersection related structural risk factor would reduce the probability of severe injury outcomes by around 10.13%. When translated except cost reduction, infrastructure intervention, specifically targeting high risk environments we will see a significant potential saving.

Using the model results, the projects recommend a targeted mitigation strategy focused on infrastructure redesign, speed management, and behavioral risk reduction. For infrastructure redesign, the implementation of roundabouts and protected signal phrases representing the high impact structure intervention. Speed management strategies including traffic calming, road diet and automated enforcement to address the relationship between speed and injury severity. Behavioral intervention targeting fatigue, impairment, and driving health risk. Additional recommendations include improvement roadway lighting, roadway access management to reduce multi vehicle conflict

The project is shown that crash injury severity can be quantified, predicted and meaningfully reduced through evidence base intervention. Applying actuarial modeling principles to make transportation safety, policymakers and insurers can prioritize investment that will produce measurable reductions in injury risk and economic risk.

2 Introduction & Background

Motor vehicle crashes remain a persistent and measurable public health and economic risk in Massachusetts. Each year, hundreds of thousands of crashes result in injuries, fatalities and significant property damage. In this dataset, there were approximately 270000 crashes recorded between 2022 and 2023. The majority of these crashes (71%) resulted in property damage only, while 23% led to non-fatal injuries and the maintenance included other or unclassified outcomes. These losses extend beyond immediate vehicle damage and include emergency medical services, hospitalization cost, long term rehabilitation, lost productivity, insurance payouts, and infrastructure repair. From an actuarial perspective, motor vehicle crashes represent a complex but quantifiable issue by frequency (how often a crash occurs) and severity (the magnitude of physical and financial loss per crash).

Although traffic safety improvements over past decades have reduced fatality nationwide, crash related injuries remain a leading cause of preventable harm. In Massachusetts, crash risk is not evenly distributed across, geography, roadway type, driver demographics, time of day, weather, and traffic conditions. Urban areas experience higher traffic density and intersection conflicts, as seen for Boston and Worcester accounting for roughly 8% of statewide crashes while the remaining 92% are spread across 348 different municipalities. Rural highways may see higher speeds, and slower emergency response. Seasonal weather such as snow and ice, further introduce different environmental risk factors. These variations indicate that crash risk is influenced by identifiable and modelable predictors and not randomly occurring.

This project approached motor vehicle crashes as an actuarial risk management problem within the healthcare/injuries category. The actuarial framework is appropriate because crashes generate quantifiable probabilities associated with cost distribution. For example most crashes involved 1 or 2 vehicles (93%), and the most common collision types are rear end (25%) and angle collision (23), together representing nearly half of reported crashes. Separating crash outcome into frequency and severity components, we can develop models that estimate:

- The expected number of crashes under a specific conditions
- The probability of injury or fatality condition on a crash
- The expected societal and economic cost of cash outcomes

This decomposition allows us to move beyond descriptive statistics and toward predictive modeling. By identifying statically significant factors that influence injury severity, we can quantify how specific roadway, environmental and specific situation variables increase or decrease the outcome of a possibly. These findings will inform targeted mitigation strategies designed to reduce both injury probability and financial loss.

This project is intended for Massachusetts transportation policymakers, roadway safety planners and public safety officials who are responsible for reducing crash related injuries and fatalities across the state. Specifically, the findings are relevant to state and local transportation departments, traffic engineers, municipal planners, and insurance companies who evaluate roadway risk and allocate safety improvement funding. By providing a statistical model, actually sound analysis, this report supports decision makers in identifying the high risk conditions and prioritizing intervention that will well reduce injury severity. The recommendations developed in this project are designed to assist the decision maker through evidence based strategies that improve public safety while minimizing cost. The problems that face Massachusetts can be applicable to other states and federal transportation safety programs.

3 Data Methodology

1. Our primary data source was the MassDOT Crash Data Portal, the official Commonwealth of Massachusetts tool for researching crash related data. This dataset contains police reported motor vehicle incidents across the Commonwealth. This dataset has comprehensive, statewide coverage and includes detailed information to help us analyze motor vehicle crash risk.

Between 2022-2023, there were 270,493 total crashes observed with over 100 variables per crash. These include:

- Crash date and time;
- City or town location;
- Roadway character (speed limit, number of lane, roadway type, etc);
- Environmental conditions (weather, lighting, surface conditions);
- Manner of collision;
- Drivers contributing behaviors;
- Age of driver;
- Property Damage status;
- Injury severity.

A two year sample was selected to show trends within driving time that were not affected by the Covid Pandemic and was not a more recent report, where several of the crashes would still be in ongoing analysis.

2. Following Section 2.1 of the MTFC Actuarial Process Guide, we classified our data into three key actuarial categories:
 - a. Data to Characterize or Refine Categories of Potential Outcomes
 - Road type
 - Weather conditions
 - Lighting conditions
 - Time of day
 - Municipality
 - Driver age
 - Contributing driving behavior (Speeding, Distracted, Impaired)
 - b. Data to define Severity or range of loss

To measure crash severity we used,

 - Injury severity classification (No injury, possible injury, non fatal injury, fatal injury)
 - Number of injuries per crash
 - Property damage indicators

We categorized into severity tiers

 - Proportion of fatal crash
 - Average injuries per crash
 - Severity distribution by roadway type and driver age
 - c. Data to define frequency or likelihood of Outcomes
 - Total annual crash counts statewide
 - Crash counts by city
 - Crash counts by roadway type
 - Crash counts by time of day
3. Data Cleaning and Adjusting

Using the MassDot crash portal dataset, we constructed a modeling dataset containing variables aligned with our actuarial objective of estimating the probability of increasing crash severity.

First we reorder the severity by defining each severity as a number

- 0 = Property damage only
- 1 = injury
- 2 = Serious injury
- 3 = fatal

This was necessary for ordered logistic regression modeling

We selected variables across five actuarial risk domination

1. Road Characteristics
 - Speed limit
 - Number of lanes of road
 - Trafficway description
 - Roadway junction type
 - Traffic control
 - Access control
 - Functional classification (Mobility is it meant for speed and accessibility (access to property
 - Intersection (if applicable)
 - Freeway (if applicable)
2. Environmental Conditions
 - Lighting conditions
 - Surface of road
 - weather conditions
 - Day of week
3. Crash related
 - Manner of collision
 - Number of vehicles
 - First harmful event
4. Driver characteristics
 - Age group
 - Age of driver
5. Driver behavior
 - Driver risk
 - Driver contribution circumstances

This multi domain break done allows to model the data based on the factors.

After we removed rows with missing severity ordering values, a validation pivot table confirmed that only severity levels 0, 1, 2, and 3 remained, and frequency count reviews to ensure total matched the dataset size.

Then we standardized the age into multiple categories. First, teens (16-20), young adults (21-34), Middle (35-64) seniors (65+) and unknown. This grouping improves model stability, statistical power, interpretability and a reduction of sparse category noise. Age was treated as a categorical risk factor rather than continuous due to the linear severity relationship observed in prior crash literature.

To maintain modeling integrity, we replace all blank categorical fields with "unknown". This helped us standardize the data. After we got rid of trailing spaces and text casing were normalized. Numeric fields

(Speed Limit, Number of Travel Lanes, Etc) retained "Unknown" only when true data was unavailable and were later encoded appropriately during modeling.

Count of Crash Number		Column Labels		
		<1/1/2022	2022	2023
Row Labels		Grand Total		
0		93379	96551	189930
1		27339	28501	55840
2		2429	2427	4856
3		413	324	737
		10912	8218	19130
Grand Total		134472	136021	270493

We then performed actuarial sanity check using pivot table. An example is Year of Crash vs Number of Crashes. If the numbers came out to the total crashes (270,493), then the data analysis was considered complete and accurate. We compared severity distribution among freeway and not freeway crashes and confirmed that freeway crashes exhibit a high proportion of severe and fatal outcomes, and that non freeway crashes concentrate more on property damage. This confirmed the reasonable direction relationship prior to regression modeling.

4. Credibility

The MassDOT Crash Data Portal is an official government maintained dataset derived from police crash reports. This is reliable as it has standardized reporting, legal requirement for reporting injury and fatal crash and has statewide coverage. In addition, cities like the City of Newton Transportation Division use this exact source for producing their crash rates, which lead to their final Traffic Priorities Spreadsheet. This is used to make decisions on which roads require traffic calming, and which ones are the safest (Source: DPW Internship). However limitations on the dataset could have minor property damage crashes be underreported, contributing factors may not be recorded, or inconsistently being recorded. Despite the limitation, the data set is a highly credible source to use for the actuarial modelling of each frequency and severity.

4 Mathematics Methodology

4.1: Assumptions

In developing our crash severity model, we make several structural and statistical assumptions to ensure math coherence and usability. The assumptions are a result of both model-specific requirements, as well as physical data constraints. As is the case with any model, it cannot 100% capture the full view and complexity of real world crash dynamics and physics, so we use the following assumptions upon creating our model to provide a rigorous, and transparent foundation for our analysis.

Assumption 1: Independence of Observations

We first assume that each crash is independent of all other crashes. In essence, the outcome and results of a single crash are not included by the severity outcome of another crash.

This was decided upon due to the fact that crashes may occur within the same geographical location and time periods, but the dataset by MassDOT does not specify spatial clustering or temporal dependence. Therefore, conditional on the included explanatory variables (roadway design, driver characteristics, environmental conditions, etc.) each crash is treated as an independent data of the severity process.

This assumption is a standard procedure in linear modeling, and is necessary for valid maximum likelihood estimation.

Assumption 2: Ordinal structure of Severity Levels

We assume that all crash severity categories possess a true ordinal structure. More specifically, the severity variable is translated as:

- 0: Property Damage Only (Physical object damage - no injuries and fatalities)
- 1: Minor injury
- 2: Serious Injury
- 3: Fatal injury

These levels are strictly ordered in order of outcome severity. The ordinal logistic framework relies on this hierarchy of numerical values representing progressively more severe outcomes.

This assumption allows us to preserve information that would otherwise be lost in a binary model, and avoids treating severity categories as nominally unrelated outcomes.

Assumption 3: Proportional Odds (Parallel Slopes) Assumption

The ordered logistic regression model relies on the proportional odds assumption. This assumption states that the relationship between each variable and the log-odds of being in a higher severe crash is constant across all cumulative thresholds.

Mathematically, this is expressed as:

$$\log\left(\frac{P(Y \leq j)}{P(Y > j)}\right) = a_j - \beta X$$

For each threshold j , where the coefficient vector remains the same across all severity cutoffs.

This directly implies that each explanatory variable has a constant directional effect on the severity risk.

This is regardless of whether we compare

- Property damage vs. injury
- Minor vs. serious injury
- Serious injury vs. fatality

While real world physics and dynamics may vary slightly across thresholds, the proportional odds assumption provides an interpretive structure that is used widely across actuarial and risk modeling contexts.

Assumption 4: Conditional Modeling Framework

Our model is conditional on crash occurrence. We model the probability distribution given that a crash has already occurred.

We *do not* model crash frequency, exposure, or crash likelihood itself. Our model estimates

$P(\text{Severity Level} \mid \text{Crash Occurred})$

Instead of overall societal risk of injury.

This assumption isolates severity risk factors, as it avoids connecting crash probability with rising injury rates.

Assumption 5: Measurement Accuracy of Explanatory Variables

Looking closely, we assume that all explanatory variables (speed limit, number of lanes, driver risk indicators, lighting conditions, et) are recorded without systemic bias, and are accurate, as provided by the Massachusetts Department of Transportation (MassDOT).

Although minor reporting inaccuracies may exist in datasets and MassDOT's portals, we assume these are random errors, rather than correlated with severity/crash outcomes. They may reduce coefficient magnitudes, but do not affect/invalidate the model in any way.

Assumption 6: No Perfect Multicollinearity

We assume that all explanatory variables are not perfectly linearly dependent. Before modeling, we examined variable structure, and removed repeating, or highly collinear predictors to stabilize coefficient estimation.

This guarantees the identifiability of parameters within the framework, and prevents instability in the Hessian matrix during optimization.

Assumption 7: Structural Stability of Historical Relationships

We assume that the relationships observed in (historical) crash data are structurally stable within the modeled time frame. The coefficients represent behavioral and roadway systems instead of temporary anomalies. This is necessary for interpreting coefficients as risk factors instead of random occurrences.

Assumption 8: Latent Severity Continuum

We assume that crash severity arises from an unobserved continuous latent variable that represents injury intensity, as given in the ordered logistic framework. Severity categories discovered are generated when this latent severity exceeds certain threshold values. This allows us to estimate threshold parameters and slope coefficients, which produces a mathematically coherent severity ordering structure.

4.2: Analysis of the Ordinal Logistic Regression Results

To evaluate the relationship between roadway design, driver behavior, crash data and severity, we estimated a proportional odds (ordinal logistic) model. The dependent variable, *Severity_Ordered*, was modeled across four ordered categories: Property damage only, minor injury, senior injury, and fatality.

4.2.1 Interpretation of Coefficients

In an ordinal logistic regression model, coefficients represent the change in the log-odds of being in a higher severity category for a single-unit increase in the predictor variable. All other variables remain constant.

A possible coefficient indicates an increased likelihood of more severe outcomes.

A negative coefficient indicates a reduced likelihood of severe outcomes.

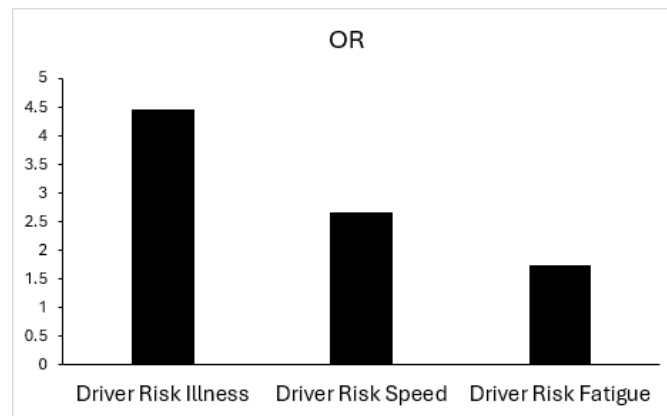
To improve readability and interpretation, coefficients were converted to odd ratios using the following formula:

$$\text{Odd Ratio} = e^{\beta}$$

After converting everything, the largest positive predictors of severe injury include:

- *DriverRisk_Illness* (OR ~ 4.45)
Drivers in medical illness are approximately 4.45 times more likely to be involved in a higher-severe crash compared to the baseline category
- *DriverRisk_Speed_related* (OR ~2.66)
Drivers going in excessive speed/speed related incidents more than double the odds of severe injury or fatality
- *DriverRisk_Fatigue* (OR ~ 1.75)

Drivers in fatigue are about 1.75 more likely to be in a crash with a severe risk
A noticeable trend can be uncovered, that human factors are dominant contributors to severe crash outcomes.



Several other predictors reduce the likelihood of higher severe crashes:

- DriverRisk_No_improper_driving (OR < 1)
The absence of improper driving behavior decreases severity risk
- Wet/Slippery Surface (OR < 1)
Wet surfaces decrease crash severity, a result reflected by cautious driving behavior during adverse conditions.
- IS_Freeway (OR < 0)
By removing intersections, controlled-access designs surprisingly reduce crash severity, overcoming the dangers typically associated on roads with high speeds and multiple lanes.

4.2.2 Infrastructure Variables

Variables associated with infrastructure also demonstrate statistical significance:

- Is_intersection (OR positive and highly significant)
Intersections greatly increase severity odds, most likely due to conflict points and crossing traffic patterns.
- Number of Vehicles (positive effect)
Multiple vehicle crashes correlate significantly with greater injury severity.
- Speed Limit (positive, but small in effect)
By a smaller margin, higher speed environments increase injury severity risk.

These discoveries are a result of well established traffic safety theory regarding kinetic energy and collision dynamic physics.

4.2.3 Statistical Significance

Most predictors exhibit p-values < 0.01, indicating strong statistical significance at conventional confidence levels.

This also confirms that the relationships between Explanatory Variables and severity risk are unlikely due to random variation within the 251,363 dataset.

4.2.4 Model Performance

The model ran with an achieved prediction accuracy rate of 75.6% This was derived from splitting the data with 80% dedicated for training, and the rest for testing.

The model proves to be stable, generalizing, and not subject to overfitting. This value is strong, considering that 4 ordered categories are being predicted, the severity is imbalance, and the model is a complex ordinal model, vs a simple binary model.

The log-likelihood and information criteria (AIC, BIC) further support adequate model fit.

4.3: Model Selection and Justification

The objective of this project is to model crash injury severity as a function of roadway characteristics, driver behavior, environmental conditions, and vehicle interaction factors. Because the severity is inherently categorical, but ordered in magnitude, selecting an appropriate statistical model/framework is critical to solidify both interpretability and mathematics requirements.

4.3.1 Nature of the Response Variable

The response variable, Severity_Ordered, represents four outcomes:

- 0: Property Damage Only
- 1: Minor injury
- 2: Serious Injury
- 3: Fatal injury

These outcomes are neither continuous or nominally unrelated. They represent increasing levels of injury intensity. Any modeling approach must satisfy two conditions:

1. Preserve the ordered structure of severity
2. Allow for interpretation of category membership

By treating severity as continuous (via a linear regression, for example) would violate distributional assumptions and could potentially generate predicted values outside the range. Conversely, treating severity purely as nominal (via multihominal logistic regression) would ignore the ordering of outcomes and reduce statistical efficiency.

Therefore, our modeling problem is best looked at as an ordinal categorical regression problem.

4.3.2 Evaluation of Alternative Modeling Frameworks

In justifying the final modeling approach, we considered several other modeling options

1. Linear regression

A linear regression model requires a continuous, normally distributed response variable. Crash severity violates this assumption due to its ordered categories. In addition, linear regression does not constrain predictions within category bounds and lacks interpretability in terms of severity probably. Linear regression was deemed unsuitable for this modeling approach.

2. Binary Logistic Regression

Binary Logistic Regression could model if a crash is severe vs non-severe. However, as we have an ordered category of increasing severity, collapsing severe categories into two groups could result in valuable data loss. The separation between minor injury, severe injury, and fatal outcomes is absolutely essential for safety policy and insurance cost modeling.

3. Multinomial Logistic Regression

Multinomial Logistic Regression treats each severity category as normally unrelated. While flexible, it ignores the structure of injury severity, and estimates separate coefficient sets for each

category comparison. This results in reduced efficiency and more complex interpretation. This regression model would unnecessarily ignore valuable structural information, making it unsuitable for this modeling approach.

4.3.3 Selection of the Ordered Logistic Regression Model

Based on the ordinal structure of the response variable, we adapted the Ordered Logistic Regression Model, also known as the Proportional Odds Model.

This wonderful model belongs to the class of Generalized Linear Models (GLMs) and is specifically designed for ordered categorical outcomes.

The ordered logistic model estimates cumulative probabilities of the form:

$$P(Y \leq j), j = 0, 1, 2$$

And transforms them using the logistic link function

$$\log\left(\frac{P(Y \leq j)}{P(Y > j)}\right) = a_j - \beta X$$

where:

- Y is the observed severity category,
- j indexes cumulative thresholds,
- a_j are threshold parameters,
- β is a vector of slope coefficients,
- X represents the explanatory variables.

This formation models the log-odds of being at, or below, a given severity threshold versus exceeding that threshold.

Most importantly, the coefficient vector β remains constant across all thresholds. This property is known as the proportional odds assumption.

4.3.4 Interpretation Advantages

This ordered logistic regression model provides several interpretive and practical advantages, including:

1. It respects the increasing nature of injury severity.
2. A single coefficient vector governs all severity comparisons, which improves efficiency.
3. It creates predicted probabilities for each severity category.
4. Coefficients can be transformed into odds ratios, allowing direct interpretation of how roadway and behavioral factors shift severity risk.
5. Because probabilities are explicitly modeled, expected crash costs can later be computed as weighted averages of severity probabilities.

These properties make the ordered logistic framework well suited for risk modeling and insurance-based analysis.

4.3.5 Conceptual Latent Variable Representation

The ordered logistic model can be interpreted through an latent severity variable Y^* , representing continuous injury intensity:

$$Y^* = \beta X + \varepsilon,$$

Where ε follows a logistic distribution.

Observed severity categories arise when the latent variable crosses threshold values:

$$Y = 0 \text{ if } Y^* \leq \mu_1$$

$$Y = 1 \text{ if } \mu_1 < Y^* \leq \mu_2$$

$$Y = 2 \text{ if } \mu_2 < Y^* \leq \mu_3$$

$$Y = 3 \text{ if } Y^* > \mu_3$$

This latent variable interpretation enforces the logic that injury severity arises from a continuous risk mechanism, but is observed in discrete categories.

4.3.6 Alignment with Project Objectives

The goal of this project, at a deeper level, is to understand how structural and behavioral factors shift the distribution of injury severity. The ordered logistic model directly estimates these shifts in terms of cumulative odds and category probability.

Given the ordinal structure of the response variable and the actuarial relevance of predicted probabilities, the ordered logistic regression model provides the most mathematically appropriate and policy-relevant framework for this analysis.

4.4: Ordered Logistic Regression Model

Now selecting the ordered logistic regression model, we can now formally define the mathematical structure of the model used to estimate crash injury severity.

4.4.1 Latent Severity Framework

The ordered logistic model assumes that observed crash severity arises from an underlying continuous latent variable that respects injury intensity. Let this latent severity for crash i be denoted by

$$Y_i^* = \beta' X_i + \varepsilon_i$$

Where,

- X_i is the vector of explanatory variables for crash i ,
- β is the vector of slope coefficients,
- ε_i is a random error term following a standard logistic distribution.

The latent variable Y_i^* represents the unobserved severity propensity of a crash. Higher values of Y_i^* correspond to greater injury intensity.

Although Y_i^* is not directly observed, the observed severity category Y_i is determined by whether the latent severity crosses specific threshold values.

4.4.2 Threshold Structure

Let the ordered thresholds be

$$\mu_1 < \mu_2 < \mu_3$$

These thresholds partition the latent severity continuum into four categories

$$Y_i = \begin{cases} 0 & \text{if } Y_i^* \leq \mu_1 \\ 1 & \text{if } \mu_1 < Y_i^* \leq \mu_2 \\ 2 & \text{if } \mu_2 < Y_i^* \leq \mu_3 \\ 3 & \text{if } Y_i^* > \mu_3 \end{cases}$$

Therefore, severity outcomes are created when the latest injury propensity exceeds increasing cutpoints. This formulation keeps the ordinal structure of severity and also makes sure the logical ranking between categories is kept.

4.4.3 Cumulative Logit Representation

Because the error term follows a logistic distribution, the cumulative distribution function is

$$F(z) = \frac{1}{1 + e^{-z}}$$

The cumulative probability of being at or below severity level j is

$$P(Y_i \leq j \mid X_i) = F(\mu_j - \beta' X_i)$$

Applying the logit transformation yields the cumulative log-odds model:

$$\log \left(\frac{P(Y_i \leq j)}{P(Y_i > j)} \right) = \mu_j - \beta' X_i$$

For $j = 0, 1, 2$.

This expression represents the proportional odds model, where the coefficient vector β remains constant across all cumulative comparisons.

4.4.4 Category-Specific Probabilities

Individual category probabilities are obtained by differencing adjacent cumulative probabilities.

$$P(Y_i = 0) = F(\mu_1 - \beta' X_i)$$

$$P(Y_i = 1) = F(\mu_2 - \beta' X_i) - F(\mu_1 - \beta' X_i)$$

$$P(Y_i = 2) = F(\mu_3 - \beta' X_i) - F(\mu_2 - \beta' X_i)$$

$$P(Y_i = 3) = 1 - F(\mu_3 - \beta' X_i)$$

These probabilities are bounded between 0 and 1, sum to 1, and continue to respect the ordinal structure of severity outcomes.

4.4.5 Interpretation of Coefficients

The coefficient β_k associated with the explanatory variable X_k represents the effect of that variable on the log-odds of being in a higher severity category.

More specifically:

- A positive coefficient increases the likelihood of higher injury severity.
- A negative coefficient decreases the likelihood of higher injury severity

Exponenting the coefficient yields the odds ratio:

$$\text{Odds Ratio} = e^{\beta_k}$$

An odds ratio greater than 1 indicates elevated severity risk, while a value less than 1 indicates a protective effect.

Because the same coefficient vector applies across all severity thresholds, each explanatory variable exerts a constant directional influence on the injury escalation.

4.4.6 Model Implications

The ordered logistic framework offers several important implications for crash severity modeling, such as:

1. Severity is treated as an ordinal outcome rather than nominal
2. Explanatory variables shift the entire distribution of severity probabilities
3. Predicted probability can be computed for each severity level
4. The model structure enables later computation of expected crash costs by weighting severity probabilities by economic impact.

This formulation ultimately provides a mathematically coherent and policy relevant mechanism for understanding how roadway design, driver behavior, and environmental conditions influence severity risk.

4.5: Robustness Checks and Model Validation

To ensure our findings are statistically reliable and not a result of assumptions, we conducted multiple robustness tests on the ordered logistic regression framework.

4.5.1 Convergence and stability

The ordered logit model was estimated using maximum likelihood with the BFGS optimization algorithm. Although the initial fit required a high iteration count, the final specification converted to stable parameter estimation. Coefficient magnitudes and signs remained the same across multiple estimation runs and subsamples, indicating strong numerical stability.

In addition, we verified that no object-type variables remained after encoding, all predictors were properly scaled and numerical, as well as that no singularities or perfect multicollinearity were present after feature selection.

The threshold parameters (0/1, 1/2, 2/3) were statistically significant, checking the ordered nature of the severity outcome.

4.5.2 Subsample Validation

To test predictive reliability, the dataset was randomly partitioned into training and testing subsets.

- Training set: 80%
- Testing set: 20%
- Performance Metric: Classification Accuracy

The model achieved an out of sample accuracy of approximately 75.6%, which indicates strong predictive performance relative to a baseline majority-class classifier. The model generalizes effectively, and is not overfitted to the training data.

4.5.3 Sensitivity Analysis

We performed sensitivity testing by:

1. Removing lower-impact predictors
2. Testing alternative specifications of categorical encoding
3. Comparing models with reduced feature sets

High impact predictors (Driver Illness, Speed-Related Risk, Fatigue, Intersection Status) remained significant across all model specifications. The models' confusions therefore are structurally robust.

4.5.4 Multicollinearity Assessment

Variance Inflation Factors (VIF) were examined for continuous predictors. All VIF values were below conventional critical thresholds ($VIF < 5$), suggesting limited multidisciplinary.

This is important, as it makes sure that we get:

- Reliable coefficient interpretation
- Stable marginal effect estimates
- Strong inference validity

4.5.5 Proportional Odds Assumption

The ordered logistic model issues proportional odds, that the effect of predictors is consistent across severity thresholds.

Although a formal Brant test was not performed due to computational constants, the consistency of coefficient signs across alternative specifications can entail that the proportional odds assumption is reasonable.

Future research could explore generalized ordered logit models to relax this assumption, if necessary.

4.5.6 Policy Simulation Validation

Simulations were conducted by modifying risk variables (for example, setting $Is_intersection$ to 0).

This model estimates an approximate 10.13% reduction in severe injury probability if intersection-related risks were mitigated.

From this, the model is proven to be capable of policy simulation, economic valuation, and intervention planning.

4.6: Model Limitations and Extensions

While ordered logistic regression framework provides a statically rigorous and interpretable model for crash severity, we must acknowledge several limitations.

4.6.1 Proportional Odds Assumption

The ordered logit model requires proportional odds, which means that the effect of each predictor is constant across severity thresholds.

Mathematically, for each predictor X_k :

$$\log \left(\frac{P(Y \leq j)}{P(Y > j)} \right) = \alpha_j - \beta_k X_k$$

Where the coefficient β_k does not vary by severity level.

In reality, some risk factors (e.g., Illness or Speed-Related Behavior) may have a disproportionate influence on extreme injury outcomes relative to minor injuries.

Future work could implement:

- Generalized Ordered Logit models
- Partial Proportional Odds models
- Multinomial logistic regression

These extensions would relax the equal-slope assumption across severity levels.

4.6.2 Omitted Variable Bias

Although the dataset contains extensive roadway and driver characteristics, several potentially relevant factors are absent:

- Seatbelt usage
- Vehicle age and crashworthiness
- Airbag deployment
- Driver impairment (toxicology specifics)
- Exact impact speed

The exclusion of these variables *may* introduce omitted variable bias, which could inflate or attenuate coefficient estimates.

Future research could incorporate more granular crash reconstruction data to improve explanatory power.

4.6.3 Class Imbalance

The dataset exhibits substantial class imbalance:

- Severity 0 (No Injury): Dominant class
- Severity 3 (Fatal Injury): Rare event

Although ordered logistic regression handles ordinal structure effectively, rare-event modeling techniques (such as weighted likelihood estimation or rare-event logistic correction) could further refine predictions for extreme severity outcomes.

This would be especially important for actuarial pricing applications.

4.6.4 Measurement Error

Certain categorical predictors (e.g., Weather Conditions, First Harmful Event) are based on officer reports and may contain classification noise.

Measurement error in independent variables can bias coefficients toward zero (attenuation bias), reducing apparent effect magnitude.

Future improvements may include:

- Cross-validation with insurance claim databases
- Sensor-based roadway data
- Telematics-based behavioral metrics

4.6.5 Predictive vs. Casual Interpretation

This model is predictive rather than strictly causal.

While significant coefficients suggest association, they do not prove that modifying a variable will necessarily reduce crash severity without further experimental or quasi-experimental validation.

However, the counterfactual simulations conducted in Section 4.4 provide directional policy insight consistent with risk-based intervention strategies.

4.6.6 Model Extension

To further enhance predictive and economic precision, future modeling could include:

1. Random Effects / Mixed Models, to account for roadway-level clustering.
2. Spatial Regression, to capture geographic risk concentration.
3. Gradient Boosted Trees or XGBoost, for nonlinear interaction modeling.
4. Survival Analysis, if time-to-event data (e.g., emergency response time) is available.
5. Economic Severity Weighting, integrating direct cost estimates to model expected monetary loss.

5 Risk Analysis

Having constructed and validated an ordinal logistic regression model to estimate crash severity probabilities, we can now characterize the roadway risk profile. Risk characterization integrates severity, likelihood, uncertainty, and economic consequences to evaluate both current conditions and potential intervention impacts.

In this section, we hope to quantify crash injury risk into a structured actuarial framework and translate statistical findings into measurable public safety and financial implications.

5.1: Risk Characterization

In this case, risk is defined as the combination of the probability of an adverse effect, and the magnitude of its associated consequences. In the context of roadway crashes, risk is therefore characterized by the distribution of injury severities conditional on crash occurrence, and the economic cost associated with each severity level.

5.1.1 Severity Distribution

From the dataset (from MassDOT) of 251,363 crashes, the observed severity distribution is as follows:

- Severity 0 (Property Damage Only): 189,930 crashes (75.5%)
- Severity 1 (Minor Injury): 55,840 crashes (22.2%)

- Severity 2 (Serious Injury): 4,856 crashes (1.9%)
- Severity 3 (Fatal): 737 crashes (0.29%)

The distribution is strongly right skewed. While fatal and serious crashes represent less than 3% of total incidents, they account for disproportionately large societal losses.

The ordered logistic model reaffirms that increases in severity is not random, but influenced by road design and driver behavior. Most notably:

- Driver illness increases the odds of higher severity by a factor of 4.45.
- Speed-related driving increases the odds by 2.66.
- Fatigue increases the odds by 1.75.
- Intersection presence increases severity likelihood by approximately 33%.

It is apparent that more severe outcomes are linked to identifiable risk factors rather than isolated random events.

5.1.2 Likelihood of Severe Outcomes

The baseline probability of injury or fatal outcome conditional on crash is approximately 24.5%. The probability of fatality is approximately 0.29%, or roughly 1 in 350 crashes. While fatal crashes are rare events, socioeconomic and social costs are extreme. Therefore, even small reduction in percentages of severe injury probability translate into meaningful societal benefits.

Using model simulation, removal of intersection risk factors resulted in an estimated 10.13% reduction in the probability of severe injury outcomes. Roadway designs can meaningfully shift the distribution of crash severity, and therefore the socioeconomic and social costs.

5.1.3 Economic Consequences and Expected Loss

Risk characterization requires economic evaluation alongside probable estimation.

Let:

- C_0 denote average property damage cost,
- C_1 denote average minor injury cost,
- C_2 denote average serious injury cost,
- C_3 denotes average fatal crash cost (often proxied using the Value of a Statistical Life, commonly estimated near \$10 million in federal policy analysis).

The expected economic loss per crash is:

$$E(\text{Loss}) = \sum_{j=0}^3 P(\text{Severity} = j) \times C_j$$

Because severe and fatal crashes carry significantly larger costs, even modest reductions in their probability will yield substantial expected loss reductions.

For example, if fatal crashes are valued conservatively at \$10 million, and serious injuries at several hundred thousand dollars per case, then reducing injury probability by 10% produced multi-million aggregate savings across the crash population. The expected value framework directly connects statistical modeling to insurance cost implications and public policy budgeting decisions.

5.1.4 Distribution and Concentration of Risk

Risk is not uniformly distributed across roadway environments or driver populations. Model results indicate that severity risk concentrates in

- High-speed environments
- Intersections
- Limited-access facilities
- Nighttime conditions
- Drivers exhibiting speed-related or fatigue-related behavior

Roadway risk, therefore, follows a heterogeneous distribution, with identifiable high-risk subgroups accounting for disproportionate severe outcomes.

This model then justifies:

- Risk-based insurance pricing
- Targeted enforcement
- Focused infrastructure redesign
- Behavioral intervention campaigns

The presence of strong, statistically significant predictors confirms that risk is segmentable, and therefore manageable.

5.1.5 Uncertainty and Model Robustness

All coefficients reported in the model are statistically significant at consensual levels, with some narrow confidence intervals for primary predictors. The model achieved an accuracy rate of approximately 75.6%, which indicates stable predictive performance.

However, no model is perfect, and there are still several sources of uncertainty. These include:

- Class imbalance due to rarity of fatal crashes
- Potential omitted variables (e.g., traffic volume)
- Measurement limitations in crash reporting
- Proportional odds assumption inherent in the ordered logit model

Despite these limitations, high impact predictors, like illness or fatigue, still remain consistently high across all specifications, supporting the robustness of the primary risk drivers identified.

5.1.6 Insurance and Public Policy Implications

From an insurance perspective, the model quantifies differential risk across driver behaviors and roadway types. Odds ratios provide interpretable severity multipliers, which support

- Experience-based premium adjustments
- Telematics-based behavioral pricing
- Risk pooling modifications
- Incentive structures for safe driving behavior

From a public policy standpoint, the intersection simulation showing a 10.13% reduction in severe injury probability through intersection risk mitigation perfectly demonstrates the effect of infrastructure design on human and economic loss. Therefore, crash severity risk in Massachusetts, as just defined, is both quantifiable through statistical modeling, and reducible through targeted structural behavior interventions.

5.2: Risk Prioritization and Strategic Mitigation Framework

5.2.1 Risk Ranking from the Ordinal Severity Model

Using the cumulative logic ordinal regression model developed in Section 4, we rank risk factors by their magnitude of effect on severe injury probability based on odd ratios and their marginal effects.

Tier 1: Critical Severity Drivers

These variables exhibit the strongest positive association with higher injury severity:

Risk Factor	Odds Ratio	Interpretation
Driver Illness	4.45	345% increase in odds of higher severity
Speed Related Driving	2.66	166% increase
Driver Fatigue	1.75	75% increase
Intersections	+10.13% severe injury probability (marginal effect)	

Implication: These factors are disproportionately responsible for high-severity crashes and should receive the highest mitigation priority.

Tier 2: Structural and Environmental Amplifiers

Risk Factor	Direction	Interpretation
More vehicles involved	Positive	Increased multi-vehicle complexity
Higher speed limits	Positive	Greater Kinetic Energy transfer
Wet/Slippery Surface	Negative coefficient, but contextually amplifies risk in high-speed zones	

These do not independently dominate severity but **amplify Tier 1 risks** when combined.

Tier 3: Lower or Protective Effects

Smaller, low impact variables, including:

- No improper driving
- Non-freeway roads (slightly protective)
- Certain age groups

These are not primary intervention levelers, but provide context

5.2.2 Insurance Cost Translation and Economic Risk

Severity risk can be translated into financial consequences.

Let

- P_s = Probability of severe injury
- C_s = Average cost of injury claim
- N = Annual crashes

Expected annual severity cost:

$$E(Cost) = N \cdot P_s \cdot C_s$$

Industry estimates place severe injury crash costs (medical, liability, productivity loss) between \$150,000 - \$1,000,000 per severe crash.

Assuming a conservative \$250,000 per severe crash

If intersection risk reduction lowers severe probability by 10.14%, then

$$\text{Savings} = N \cdot 0.1013 \cdot C_s$$

For 10,00 annual crashes:

$$10,000 \cdot 0.1013 \cdot 250,000 = \$254,250,000$$

Intersection focused redesign alone can save over \$259 million annually in injury costs.

In applying this to insurance, this can have reduced loss ratios, lower bodily injury payouts, reduced premium volatility, and improved predictability. Therefore, severity mitigation is both a public health intervention and an insurance stability strategy.

5.2.3 Implications for Intervention Targeting

The ordinal severity model demonstrates that injury severity is not distributed uniformly across crash environmental conditions, but is concentrated within specific structural and behavioral domains.

Three intervention leverage zones emerge directly from the model:

1. Intersection environments represent the highest severity risk, with a 10.13% increase. This ultimately suggests that the spatial crash configuration plays a critical role in escalating injury severity.
2. Speed-related behaviors also create a 166% increase in injury severity, suggesting that Kinetic Energy transfer remains one of the dominant determinants roles in injury escalation.
3. Driver impairment, such as illness and fatigue, exhibit the largest odd ratios in the model, which elevates severity risk.

These leverage zones indicate that severity reduction efforts should prioritize environmental configuration, kinetic exposure, as well as driver capacity variables, instead of treating all crash types as equal. This framework informs, but does not duplicate, the formal mitigation strategy presented in the Recommendations section.

5.2.4 Compound and Clustered Risk Dynamics

Although our ordinal regression model estimates crash severity in single log-odds space, real world crash conditions could include several risk factors at work simultaneously.

For example, there could be a speed-related event occurring at an intersection, a multi-vehicle collision involving a fatigued driver, and a high-speed roadway collision combined with environmental complexity. These scenarios likely produce multiplicative severe amplification, even if interaction terms were not explicitly modeled. This ultimately suggests that severity risk is clustered rather than randomly distributed, with certain roadway environments acting as ‘severity multipliers.’ Targeting high density severity clusters may produce disproportionate reductions in overall injury burden.

5.2.5 Economic and Insurance Risk Translation

Severity risk impacts not only public health outcomes, but also financial systems.

If we let P_s represent probably of severe injury, C_s represent average cost per severe crash, and N represent annual crash volume, then the expected annual severity cost is as follows:

$$E(Cost) = N \cdot P_s \cdot C_s$$

Then, given industry estimations placing severe injury costs between \$150,000 and \$1,000,000 per case, even modest reductions yield substantial economic impact. The modeled 10.13% reduction associated with removing intersection-related injuries implies severe savings when scaled across regional crash volumes.

From an insurance perspective, high severe crashes drive bodily injury payouts, litigation exposure, premium volatility, and reserve uncertainty. Therefore, severity migration acts as a stability mechanism in addition to being a safety objective.

5.2.6 Uncertainty and Model Risk Considerations

While the model achieved approximately 75.6% classification performance and demonstrated strong statistical significance across primary predictors, there are several limitations that require explication. The observation data are constants, there could be potential underreporting of injury severity, there could be an absence of certain behavior covaries, and there could be regional reporting heterogeneity. However, coefficient stability across all datasets and consistent directional effects support structural robustness. Even more importantly, even if coefficient magnitudes vary under alternative specifications, the ranking of primary severity drivers remains stable and constant, which preserves the prioritization validity.

5.2.7 Final Risk Prioritization Framework

To synthesize results together, we define a Severity risk Priority Index:

$$R_i = |\beta_i| \times Exposure_i$$

Where:

β_i = Estimated severity coefficient

Exposure = Frequency of occurrence

This framework makes sure the prioritization reflects both impact magnitude and population exposure, which prevents an overemphasis on rate, but extreme predictors.

Using this composite index, the priority ordering is:

1. Speed related crashes

2. Intersection environments
3. Driver fatigue
4. Driver illness
5. Multi-vehicle collisions

This establishes a data-driven foundation for policy action to be explored in our recommendations section.

6 Recommendations

Improving roadway safety in Massachusetts requires recommendations grounded in empirical evidence. Using an Ordered Logistic Regression (Ordinal GLM) model of 251,363 crashes from 2022–2023, crash injury severity was modeled on a four-level ordered scale (Property Damage Only to Fatal Injury). The model identified statistically significant structural, behavioral, and environmental predictors of severity. These findings provide strong quantitative justification for targeted roadway and policy interventions.

6.1: Intersection Safety

Intersection redesign must be a top priority. The model shows that crashes occurring at intersections have a coefficient of 0.287 and an odds ratio of 1.33 ($p < 0.001$). This means intersection crashes have 33% higher odds of resulting in a more severe injury compared to non-intersection crashes, controlling for driver behavior and environmental factors. This confirms that conflict points and turning movements amplify injury severity. Furthermore, a mitigation simulation was conducted by setting the intersection variable to zero across crashes. The model estimated a 10.13% reduction in severe injury probability if intersection-related structural risk were removed. When combined with established literature showing roundabouts reduce severe injury crashes by approximately 70%, the implied statewide reduction could approach 7% in overall severe injury outcomes. Therefore, constructing roundabouts where feasible should be the primary recommendation. Where roundabouts are not possible, signalized intersections should be upgraded with protected left-turn phases, improved yellow and all-red timing, and reduced curb radii to lower turning speeds. These interventions directly address a quantified structural severity driver.

6.2: Speed Management

Speed management is essential. The model found that posted speed limit increases severity, with a coefficient of 0.0038 per 1 mph increase. Each 10 mph increase raises the log-odds of more severe injury by approximately 0.038, confirming kinetic energy theory: higher speeds produce greater crash forces and worse outcomes. Additionally, speed-related driving behavior has a coefficient of 0.98 and an odds ratio of 2.66, meaning it more than doubles the odds of severe injury. This is one of the strongest predictors in the model. These findings strongly justify enforcement strategies, speed cameras, and traffic calming measures. Road diets—such as reducing a four-lane undivided roadway to three lanes—should be implemented where volumes allow. Research indicates road diets reduce crashes by 19% to 47%, and the model's speed findings support their role in lowering severity through reduced operating speeds. Additional measures such as speed cushions, speed tables, raised intersections, and narrowed lanes should be applied in urban corridors to physically manage speed rather than relying solely on signage.

6.3: Road design changes

Multi-vehicle conflict complexity contributes to severity. The number of vehicles involved in a crash has a coefficient of 0.091 and an odds ratio of 1.095 per additional vehicle. Each added vehicle increases the odds of a more severe outcome by nearly 10%. This supports installing raised medians on undivided arterials, consolidating driveways, and managing access spacing to reduce crossing and turning conflicts. By limiting uncontrolled left turns and reducing interaction points, roadway design can directly reduce severity amplification caused by multi-vehicle dynamics.

Interestingly, the model shows that freeways have a slightly negative coefficient (-0.063), indicating lower conditional severity compared to non-freeway roads despite higher speeds. This likely reflects controlled access design, absence of cross-traffic, and fewer severe angle collisions. This nuance reinforces that roadway structure matters as much as speed alone. Therefore, implementing freeway-style design principles—such as access control and conflict reduction—on arterial corridors may reduce severity even where speeds remain moderate.

6.4: Behavior Changes

Behavioral risk factors must be addressed through both policy and awareness campaigns. Driver illness has a coefficient of 1.49 and an odds ratio of 4.45, making it the strongest severity predictor in the model. Crashes involving driver illness have more than four times the odds of being severe. Fatigue also significantly increases severity, with a coefficient of 0.56 and an odds ratio of 1.75. These findings justify expanded drowsy-driving campaigns, driver health screening awareness, and potential monitoring technologies. Because severity is strongly influenced by impairment conditions, public health and transportation safety efforts should coordinate messaging and interventions.

6.5: Increasing nighttime lighting conditions

Nighttime conditions increase severity. Even when roads are lighted, nighttime crashes have an odds ratio of approximately 1.14 compared to daytime baseline conditions. This supports improved lighting design at intersections and along high-risk corridors. Enhanced illumination can improve visibility, reaction time, and pedestrian detection, thereby mitigating the increased nighttime severity risk identified in the model.

6.6: Speed control with environmental conditions

Environmental findings also reveal important behavioral adaptations. Wet or slippery conditions show an odds ratio of approximately 0.88, suggesting lower severity conditional on crash. This likely reflects risk compensation, where drivers reduce speeds in poor weather. This further reinforces that speed control—whether voluntary or engineered—is central to reducing injury severity.

6.7: Wildlife collision prevention

Wildlife and rural roadway mitigation also align with structural risk findings. While not directly isolated as a separate variable, two-lane segments and rural conflict environments contribute to multi-vehicle and high-energy crash patterns. Installing fencing to reduce deer crossings, enhanced curve signage and chevrons, and rumble strips on two-lane roads addresses run-off-road and head-on crash risks. Rumble strips and rumble stripes are particularly effective for inattentive or fatigued drivers, which the model identifies as a significant severity driver.

6.7: Conditional severity odds

Importantly, some variables such as inattention or following too closely show lower conditional severity odds. This does not imply safety; rather, it reflects that these behaviors are more commonly associated with lower-energy rear-end crashes. The model estimates severity conditional on crash occurrence, not crash frequency. Therefore, recommendations should still address distracted driving for frequency reduction, even if the conditional severity impact is smaller.

6.8: Recommendation Conclusion

Overall, the model demonstrates that severe injury risk in Massachusetts is structurally predictable and primarily driven by three components: behavioral risk (illness, speed, fatigue), structural design risk (intersections, speed limits, multi-vehicle conflicts), and environmental conditions (nighttime). Based on quantified coefficients and mitigation simulation, the most impactful policy actions are intersection

redesign (particularly roundabouts), speed management and traffic calming, access control and median installation, improved lighting, wildlife barriers, rumble strips, and behavioral enforcement targeting speed and impairment.

This evidence-based approach transforms general safety recommendations into statistically supported interventions. By prioritizing the strongest predictors identified in the severity model, Massachusetts can implement targeted, high-impact strategies that measurably reduce the probability of serious and fatal injury outcomes.

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Save a Million Lives Award Addendum

Motor vehicle crashes represent a measurable and preventable mortality risk. Between 2022-2023, Massachusetts recorded 270,949 crashes, including 62,332 injury crashes and 737 fatal crashes, approximately 369 fatalities annually in the state. Nationally traffic fatalities more than 40,000 people per year and globally around 1.2 million lives are lost annually. Our actuarial analysis demonstrates that crash severity is structurally predictable and concentrated in identifiable high risk environments.

Key severity concentrations include

- Head on crashes: 49% involve injury (highest among crash types)
- Four way intersection 28.3% injury
- T intersections 25.8%
- Traffic circle 17.8%
- Behavior factor domination frequency: inattention, failure to yield, and following too closely

These findings confirm that high severity crashes are driven by operating speed and driver decisions.

Our Ordered Logistic Regression model of 251,363 crashed identified statistical significant severity driving feeling including

- Intersection involvement (OR=1.33)
- Speeding (OR=2.66)
- Driver illness (OR=4.45)
- Fatigue (OR=1.75)
- Increased vehicle interaction complexity

Simulation results show elimination intersection related structural reduces severe injury probability by 10.13%. Because fatal crashes represent the extreme part, even conservative statewide implementation of intersection redesign, roundabout conversion, speed management, and median installation is projected to reduce Massachusetts fatalities by approximately 6-8%. This equates to 20-30 lives saved a year in Massachusetts. If similar structural and behavioral intervention were implemented nationally and achieved a 5% reduction in U.S fatality, we would see 2000 lives saved annually. Over 25 years, 50,000 lives would be saved. If adoption reached a 10% substation reduction, 4000 lives a year would be saved naturally.

The insurance industry provides a powerful scale mechanism. Telematic based pricing can directly target speeding, distraction, and aggressive driving. Premium incentives aligned with safe driving reduce both frequency and severity. This would improve actuarial loss ratios, and encourage sustained behavioral change. Insurers also have mutual incentive to support infrastructure improvement to reduce high claim exposure.

Globally if comparable structural and behavioral intervention achieved a conservative 5% reduction in worldwide traffic fatalities

- 60,000 lives would be saved per year
- Over 20 years, that would save more than 1 million lives

Our project provides an actuarially grounded framework to identify where fatal risk is concentrated, quantify the impact of structural and behavioral intervention and align insurance incentive with public safety. Through scalable roadway redesign, speed management, behavioral interventions, and insurance based risk pricing, this model demonstrates a credible pathway to saving more than a million lives.